AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM291
Management Information Systems

Final Examination
Fall Semester 2024/2025

Date: .9/Jan./2025

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Number of Exam Pages: 4 (including this cover sheet)	**Time Allowed:** 2 Hours

Instructions:

- Total Marks: **60**
- This exam consists of **3** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are **not** allowed.

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PART 1: Multiple Choice Questions – [10 Marks]

Question 1: [10 Marks]

This part consists of 10 questions carrying a WEIGHT OF 1 mark for each. Answer the questions by filling in the blanks with the correct word/expression.

1. A ..... is an organized collection of data.
2. A ..... is a software package that can be used for creating and managing databases.
3. A .....is a unique value that identifies a row in a table.
4. A .....is a set of data elements that is organized using a model of vertical columns and horizontal rows.
5. A ..... represents a single, data item in a table.
6. ..... command is used to create a table in SQL.
7. A ..... is a standard for commands that define the different structures in a database.
8. A .....is a language that enables users to access and manipulate data in a database.
9. ..... are the basic building blocks of a database.
10. There are ..... types of Relationships in relational model.

PART 2: Short Questions – [20 Marks]

Question 2: [10 Marks]

Draw an ER diagram for the following scenario:

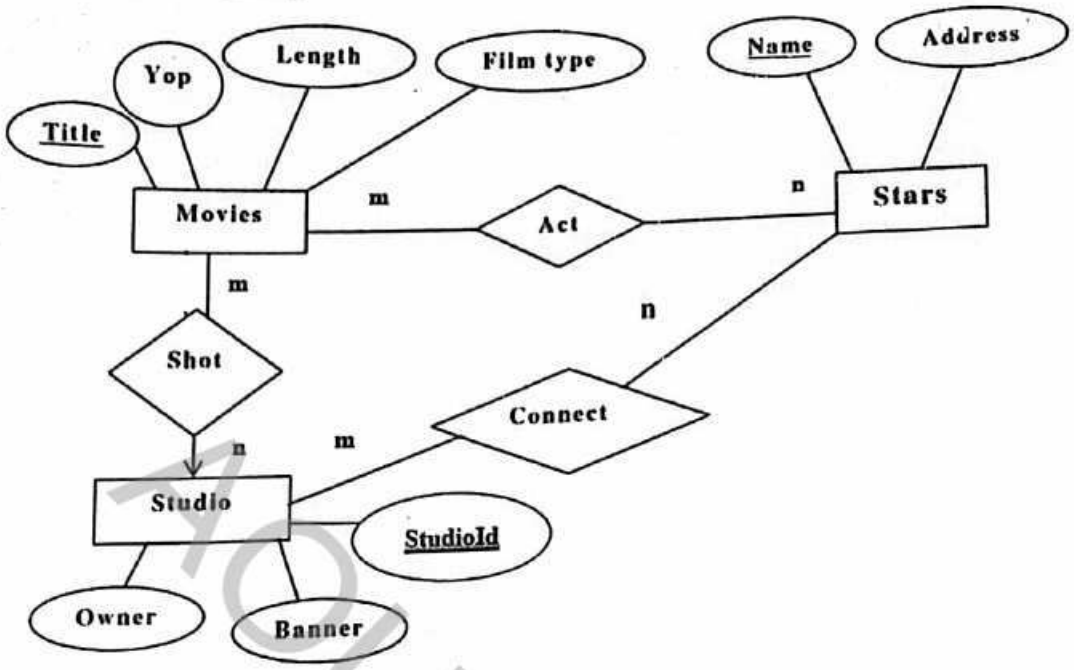
Set up a database company, ArtBase that builds a product for art galleries. The core of this product is a database with a schema that captures all the information that galleries need to maintain.

- Galleries keep information about artists, their names (which are unique), birthplaces, age, and style of art.
- For each piece of artwork, the artist, the year it was made, its unique title, its type of art (e.g., painting, lithograph, sculpture, photograph), and its price must be stored.
- Pieces of artwork are also classified into groups of various kinds, for example, portraits, still life's, works by Picasso, or works of the 19th century; a given piece may belong to more than one group.
- Each group is identified by a name (like those just given) that describes the group.
- Finally, galleries keep information about customers. For each customer, galleries keep that person's unique name, address, total amount of dollars spent in the gallery (very important!), and the artists and groups of art that the customer tends to like.

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Question 3: [10 Marks]
Map the following ERD into a relational model. Identify the primary key for each relation and the necessary foreign keys.



PART 3: Long Questions – [30 Marks]

Answer ALL the following questions

Question 4 [10 Marks]
Study the following database table that stores transaction records for bank clients and answer the questions that follow.

TABLE : ACCOUNT

ANO	ANAME	ADDRESS
101	Nirja Singh	Bangalore
102	Rohan Gupta	Chennai
103	Ali Reza	Hyderabad
104	Rishabh Jain	Chennai
105	Simran Kaur	Chandigarh

TABLE: TRANSACT

TRNO	ANO	AMOUNT	TYPE	DOT
T001	101	2500	Withdraw	2017-12-21
T002	103	3000	Deposit	2017-06-01
T003	102	2000	Withdraw	2017-05-12
T004	103	1000	Deposit	2017-10-22
T005	102	12000	Deposit	2017-11-06

a. Display details of all transactions of type withdraw.

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b. Display ANO and AMOUNT for all transactions done in 12th of 'May' 2017.
c. Display date of transaction done for account having ANO as 102.
d. Display the names for all clients who have transactions in TRANSACT table.
e. Display the total amount for all transactions done before October month 2017.

Question 5 [10 Marks]

In a distributed database environment, the following query (Q) is submitted and processed at site 3 against tables Student (at site 1) and Registration (at site 2):

Student: (1,000 rows. Row size = 50 bytes. Table size = 50,000 bytes.)

Name	SID	Bdate	Address	Sex	GPA

Registration: (1,000 rows. Row size = 30 bytes. Table size = 30,000 bytes.)

SID	CCode	Grade

Q: For each student, retrieve student name and grade for each registered course.
Suppose that the result of this query is 5,000 tuples and each tuple is 20 bytes long. Discuss three strategies and calculate the cost of transferring data over the network, then decide which strategy is the optimal.

Question 6 [10 Marks]

Consider the manufacture schema below. Manufacture consists of tables Suppliers, Parts, Project and SPJ relation instances in the relations below.

S

SNO	SNAME	CITY
S1	Smita	Delhi
S2	Jim	Pune
S3	Ballav	Pune

P

PNO	PNAME	COLOUR	CITY
P1	Nut	Red	Delhi
P2	Bolt	Blue	Pune
P3	Pen	White	Mumbai

J

JNO	JNAME	CITY
J1	Sorter	Pune
J2	Display	Mumbai

SPJ

SNO	PNO	JNO	QUANTITY
S1	P1	J1	200
S2	P2	J2	700

What would be the constraint violations if the following operations are performed:

(a) Insert <Null, 'jack', 'Mumbai'> into S (2 marks)
(b) Insert <'S2', Null, 'J3', 200> into SPJ (2 marks)
(c) Insert <'P2', 'Pencil', 'Grey', 'Kolkata'> into P (2 marks)
(d) Insert <'J3', 'Monitor', 'Jaipur'> into J (2 marks)
(e) Update tuple1 in S relation to <Null, 'Smith', 'Pune'> (2 marks)
(f) Delete tuple1 in J (2 marks)
(g) Update tuple2 in SPJ to <'S3', 'P2', 'J3', 800> (2 marks)
(h) Delete tuple1 in SPJ. (2 marks)
(i) Update tuple3 in S to <'S3', 1, 'Mumbai'> (2 marks)
(j) Update tuple1 in SPJ to <'S1', Null, 'J1', 300> (2 marks)

End of questions

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